the plates of the capacitor (gap between the plates). So the magnetic field at a point P is zero. Hence there is an inconsistency between equations (5.3) and (5.4).

Maxwell resolved this inconsistency as follows: While the capacitor is being charged up, varying electric field is produced between capacitor plates. There must be a current associated with the changing electric field between capacitor plates. In other words, time–varying electric field (or time–varying electric flux) produces a current. This is known as displacement current flowing between the plates of the capacitor (Figure 5.5).

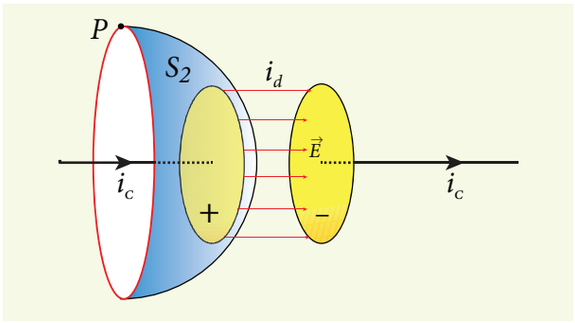


Figure 5.5 Applying Gauss's law between the plates of the capacitor

From Gauss's law of electrostatics, the electric flux between the plates of the capacitor is

$$\Phi_E = \oint_S \vec{E}\cdot d\vec{A} = EA = \frac{q}{\epsilon_\circ}$$

where A is the area of the plates of capacitor. The change in electric flux is given by

$$\frac{d\Phi_E}{dt} = \frac{1}{\epsilon_o}\frac{dq}{dt} \quad \text{(or)}$$

$$\frac{dq}{dt} = \epsilon_o \frac{d\Phi_E}{dt}$$

$$i_d = \epsilon_o \frac{d\Phi_E}{dt} \qquad (5.5)$$

where $\frac{dq}{dt} = i_d$ is known as displacement current or Maxwell's displacement current.

The **displacement current can be defined as the current which comes into play in the region in which the electric field (or the electric flux) is changing with time**. In other words, whenever the change in electric field takes place, displacement current is produced.

Maxwell modified Ampere's law as

$$\oint_l \vec{B}.d\vec{l} = \mu_0 i = \mu_0 [i_c + i_d]$$

$$\oint_l \vec{B}.d\vec{l} = \mu_0 i_c + \mu_0 \epsilon_0 \frac{d\Phi_E}{dt} \qquad (5.6)$$

where the total current enclosed by the surface becomes the sum of conduction current and displacement current. Therefore, $i = i_c + i_d$. The equation (5.6) is known as **Ampere–Maxwell law**. When the current in the circuit is constant, the displacement current is zero.

Between the plates, the conduction current is zero while the displacement current is non–zero. This displacement current or time–varying electric field can also produce a magnetic field between the plates of the capacitor. The magnetic field at a point inside the capacitor is perpendicular to the electric field and is shown in Figure 5.6. This magnetic field can be determined using equation (5.6).

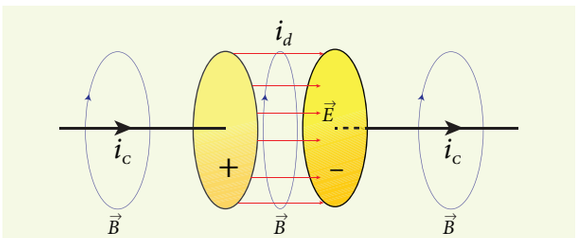


Figure 5.6 Magnetic field produced by conduction and displacement currents

UNIT-5(XII-Physics_Vol-1).indd 269 14-02-2020 20:11:18